

NTG

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \theta > 1$$

a) OMT

$$L(\theta) = \prod_{i=1}^n \frac{\theta-1}{x_i^\theta} = (\theta-1)^n \cdot \prod_{i=1}^n \frac{1}{x_i^\theta} =$$

$$= (\theta-1)^n \left(\prod_{i=1}^n \frac{1}{x_i} \right)^\theta$$

$$\ln L(\theta) = n \ln(\theta-1) + \theta \ln \prod_{i=1}^n \frac{1}{x_i}$$

$$\frac{\partial}{\partial \theta} (\ln L(\theta)) = \frac{n}{\theta-1} + \ln \prod_{i=1}^n \frac{1}{x_i}, \text{ t.e.}$$

$$\theta = 1 + n \cdot \frac{1}{-\ln \prod_{i=1}^n \frac{1}{x_i}} = 1 + \frac{n}{\ln \prod_{i=1}^n x_i}$$

$$\text{Hence } \theta_0 = 1 + \frac{n}{\sum_{i=1}^n \ln x_i}$$

$$\frac{\partial^2}{\partial \theta^2} (\ln L(\theta)) = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow \theta_0 - \text{max}$$

$$\Rightarrow \tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^n \ln x_i}$$

b) m-мерная

$$\frac{1-\theta-1}{x} = \frac{(1-\theta)x^{(\theta-1)}}{x}$$

$$y = \int_{-\infty}^m p(x) dx = \frac{1}{2}$$

$$y = \int_1^m \frac{\theta-1}{x^\theta} dx = (\theta-1) \left(\frac{1}{(1-\theta)x^{(\theta-1)}} \right) \Big|_1^m =$$

$$= 1 - \frac{1}{m^{\theta-1}} = \frac{1}{2}$$

$$\Rightarrow m^{\theta-1} = 2 \Rightarrow m = 2^{\frac{1}{\theta-1}}$$

Необх. проверить модель на сильную
рекуррентность.

Рекуррентность:

1. $F(x, \theta)$ - непрерыв. дифф. по θ на $\mathbb{R}$

$$2. \frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} f = 0$$

$$\begin{aligned} \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) &= \int_1^{+\infty} \frac{\partial p}{\partial \theta} dx = \int_1^{+\infty} \frac{x^\theta - (\theta-1)x^\theta \ln x}{x^{2\theta}} dx = \\ &= \int_1^{+\infty} \frac{1}{x^\theta} dx - (\theta-1) \int_1^{+\infty} \frac{\ln x}{x^\theta} dx \end{aligned}$$

$$\frac{\partial F}{\partial \theta} = \frac{\partial}{\partial \theta} \left(1 - \frac{1}{x^{\theta-1}} \right) = -\frac{\partial}{\partial \theta} x^{(1-\theta)} = x^{(1-\theta)} \ln x$$

$$\int_1^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \left. \frac{\ln x}{x^{\theta-1}} \right|_1^{+\infty} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^{\theta-1}} \stackrel{\theta > 1}{=} 0$$

$$\Rightarrow \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF = \int_{-\infty}^{+\infty} d\left(\frac{\partial}{\partial \theta}\right)$$

$$3) I(\theta) = \int_{A(\theta)} \left(\frac{\partial \ln p}{\partial \theta} \right)^2 p dx = \int_1^{+\infty} \left(\frac{\partial}{\partial \theta} [\ln(\theta-1) - \ln x^{\theta}] \right)^2 p dx$$

$$= \int_1^{+\infty} \left(\frac{1}{\theta-1} - \ln x \right)^2 \cdot \frac{\theta-1}{x^{\theta}} dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-1} + \ln^2 x \right) \frac{\theta-1}{x^{\theta}} dx = \left\{ x = e^t \right\}$$

$$= \int_0^{+\infty} \left(\frac{1}{(\theta-1)^2} e^{-\theta t} - \frac{2t}{e^{\theta t}} + \frac{(\theta-1)t^2}{e^{\theta t}} \right) e^t dt =$$

$$= \underbrace{\int_0^{+\infty} \frac{1}{\theta-1} e^{-t(\theta-1)} dt}_{W_1} - \underbrace{\frac{2}{1} \int_0^{+\infty} t e^{-t(\theta-1)} dt}_{W_2} + \underbrace{(\theta-1) \int_0^{+\infty} t^2 e^{-t(\theta-1)} dt}_{W_3}$$

$$W_1 = \frac{1}{(\theta-1)^2}$$

$$W_2 = \frac{1}{(\theta-1)^2}$$

$$W_3 = 2 \frac{1}{(\theta-1)^2}$$

$$\Rightarrow I(\theta) = \frac{1}{(\theta-1)^2} \quad - \text{кеир. на } \theta \text{ координате}$$

$$\text{из } \theta > 1$$

$\Rightarrow$ модель регрессии

$$F = 1 - \frac{1}{x^{\theta-1}} \cdot \frac{\partial F}{\partial \theta} = \frac{\ln x}{x^{\theta-1}} \quad - \text{кеир. на } \theta > 1$$

$$\frac{\partial^2 F}{\partial \theta^2} = \frac{\ln^2 x}{x^{\theta-1}} \quad - \text{кеир. на } \theta > 1$$

$\Rightarrow F$ — зрн кеир график. на $\theta > 1$

$$\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF = 0$$

$$\int_{-\infty}^{+\infty} d \frac{\partial^2 F}{\partial \theta^2} = \int_1^{+\infty} d \left(\frac{\ln^2 x}{x^{\theta-1}} \right) = \frac{\ln^2 x}{x^{\theta-1}} \Big|_1^{+\infty} = 0 = \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF$$

$\Rightarrow$ Модель сильно регрессивна
+ $\theta > 1$ — открытое ин-во.

Значит выполняем достаточное
условие асимпт. св-ва ОМТ:

$$\sqrt{n}(\theta^2 - \theta) \rightsquigarrow N(0, \frac{1}{I(\theta)}) = N(0, (\theta-1)^2)$$

Пусть $m(x) = 2^{\frac{1}{x-1}}$ — группа в $\mathbb{Q}$,

$$m'(0) = 2^{\frac{1}{x-1}} \ln 2 \cdot \left(\frac{-1}{(x-1)^2} \right) \Big|_{x=0} \neq 0 \quad \text{так } 0 > 1$$

$\Rightarrow$ Верно: $\sqrt{n} (m(\tilde{\theta}) - m(0)) \rightsquigarrow N(0, m'(0) \frac{1}{I(0)})$

$$\begin{aligned} \sigma(0) &= +2^{\frac{1}{0-1}} \ln 2 \cdot \frac{1}{(0-1)^2} \cdot (0-1) = \\ &= + \frac{\ln 2 \cdot 2^{\frac{1}{0-1}}}{0-1} \end{aligned}$$

$$\sqrt{n} \frac{m(\tilde{\theta}) - m(0)}{\sigma(\tilde{\theta})} \rightsquigarrow N(0, 1)$$

$$t_1 = U_{\frac{1-\beta}{2}}; t_2 = U_{\frac{1+\beta}{2}}$$

$$t_1 = \sqrt{n} \frac{m(\tilde{\theta}) - m(0)}{\sigma(\tilde{\theta})} < t_2$$

$$m(\tilde{\theta}) - \frac{\sigma(\tilde{\theta})}{\sqrt{n}} t_2 < m < m(\tilde{\theta}) - \frac{\sigma(\tilde{\theta})}{\sqrt{n}} t_2$$

c) $\sqrt{n} (\tilde{\theta} - 0) \rightsquigarrow N(0, \frac{1}{I(0)})$

$$\frac{\sqrt{n} (\tilde{\theta} - 0)}{\sqrt{\frac{1}{I(0)}}} \rightsquigarrow N(0, 1)$$

$$\tilde{\theta} - \frac{t_2}{\sqrt{n I(\tilde{\theta})}} < 0 < \tilde{\theta} - \frac{t_1}{\sqrt{n I(\tilde{\theta})}}$$

d), t), f) —
— см. 6 параграфа
"d-t-f points.
ipy nb"